Lab 8

2026-04-03

```r
suppressMessages(library(DBI))
```

```
## Warning: package 'DBI' was built under R version 4.4.3
```

```r
suppressMessages(library(odbc))
```

```
## Warning: package 'odbc' was built under R version 4.4.3
```

```r
suppressMessages(library(RMySQL))
suppressMessages(library(dplyr))
suppressMessages(library(tidyverse))
```

```
## Warning: package 'tidyr' was built under R version 4.4.3
```

```r
suppressMessages(library(dbplyr))
```

```r
db_user <- 'studentUser'
db_password <- 'STAread!'
db_name <- 'Test'
db_table <- 'your_data_table'
db_host <- 'aiken.cs.grinnell.edu'
db_port <- 3306
mydb <- dbConnect(MySQL(), user = db_user, password = db_password,
                  dbname = db_name, host = db_host, port = db_port)
```

Question 1

```r
dbListTables(mydb)
```

```
##  [1] "course"           "course_offering"  "customers"        "office_hours"
##  [5] "orders"           "professor_info"   "syllabus_info"    "week9courses"
##  [9] "week9departments" "week9profcourses" "week9professors"
```

Question 2

```r
# return the first 3 rows of the four week9 tables
dbGetQuery(mydb, "SELECT * FROM week9courses
           LIMIT 3")
```

```
##      cid                      cname did
## 1 CHM129    General Chemistry w/lab CHM
## 2 CHM210 Analytical Chemistry w/Lab CHM
## 3 CHM221  Organic Chemistry I w/lab CHM
```

```r
dbGetQuery(mydb, "SELECT * FROM week9departments
           LIMIT 3")
```

```
##   did          depname
```

```
## 1 CHM        Chemistry
## 2 CSC Computer Science
## 3 ECN        Economics
```

```r
dbGetQuery(mydb, "SELECT * FROM week9profcourses
           LIMIT 3")
```

```
##   pid    cid
## 1  42 MAT131
## 2  42 MAT218
## 3  41 ECN111
```

```r
dbGetQuery(mydb, "SELECT * FROM week9professors
           LIMIT 3")
```

```
##   pid    fname      email did
## 1  42    Royce    [Royce] MAT
## 2  41 Abhinaba [Abhinaba] ECN
## 3  40   Andrea   [Andrea] ECN
```

- week9courses represent courses id, course name, and department id of all courses in week 9
- week9departments represent department id and department name of all courses in week 9
- week9profcourses represent professor id and courses id of all courses in week 9
- week9professors represent professor id, faculty name, their email, and department id of all courses in week 9

Question 3

```r
# store tables as dataframes
week9courses <-dbGetQuery(mydb, "SELECT * FROM week9courses")
week9departments <- dbGetQuery(mydb, "SELECT * FROM week9departments")
week9profcourses <- dbGetQuery(mydb, "SELECT * FROM week9profcourses")
week9professors <- dbGetQuery(mydb, "SELECT * FROM week9professors")
```

Question 4

Part A

```r
# add department names to professor dataset sql
dbGetQuery(mydb, "SELECT a.*, b.depname FROM week9professors as a
           LEFT JOIN week9departments as b ON a.did = b.did
           LIMIT 5;")
```

```
##   pid     fname       email did   depname
## 1  39    Andrew    [Andrew] CHM Chemistry
## 2  33    Corasi    [Corasi] CHM Chemistry
## 3  31    Elaine    [Elaine] CHM Chemistry
## 4  30 Elizabeth [Elizabeth] CHM Chemistry
## 5  28     Erick     [Erick] CHM Chemistry
```

Part B

```r
# add department names to professor dataset r
mytable <- left_join(x = week9professors, y = week9departments, by = "did")
```

```r
head(mytable)
```

```
##   pid    fname      email did          depname
## 1  42    Royce    [Royce] MAT      Mathematics
## 2  41 Abhinaba [Abhinaba] ECN        Economics
## 3  40   Andrea   [Andrea] ECN        Economics
## 4  39   Andrew   [Andrew] CHM        Chemistry
## 5  38  Bradley  [Bradley] ECN        Economics
## 6  37  Charlie  [Charlie] CSC Computer Science
```

Part C

- The resulting tables are the same, but they seem to be ordered differently. They include the same information because they are both left joining the same datasets.

Question 5

Part A

```r
# table including only pid and did
data <- subset(week9professors, select = c(pid, did))
head(data)
```

```
##   pid did
## 1  42 MAT
## 2  41 ECN
## 3  40 ECN
## 4  39 CHM
## 5  38 ECN
## 6  37 CSC
```

Part B

```r
# number of faculty in each department by department id
data %>% group_by(did) %>% summarize(Count = n())
```

```
## # A tibble: 5 x 2
##   did   Count
##   <chr> <int>
## 1 CHM      11
## 2 CSC       8
## 3 ECN       9
## 4 MAT       9
## 5 STA       5
```

Question 6 (Question 5 in SQL)

Part A

```r
# table containing only professor ids and department ids
dbGetQuery(mydb, "SELECT pid, did FROM week9professors
           LIMIT 5")
```

```
##   pid did
## 1  42 MAT
```

```
## 2  41 ECN
## 3  40 ECN
## 4  39 CHM
## 5  38 ECN
```

Part B

```r
# table containing only professor ids and the name of the department they are in
dbGetQuery(mydb, "SELECT pid, depname FROM week9professors as a
           LEFT JOIN week9departments as b ON a.did = b.did
           LIMIT 5")
```

```
##   pid   depname
## 1  39 Chemistry
## 2  33 Chemistry
## 3  31 Chemistry
## 4  30 Chemistry
## 5  28 Chemistry
```

Part C

```r
# number of faculty in each department by department id
dbGetQuery(mydb, "SELECT did, COUNT(*) FROM week9professors
           GROUP BY did")
```

```
##   did COUNT(*)
## 1 CHM       11
## 2 CSC        8
## 3 ECN        9
## 4 MAT        9
## 5 STA        5
```

Part D

```r
# number of faculty in each department by department name
dbGetQuery(mydb, "SELECT depname, COUNT(*) FROM week9professors as a
           LEFT JOIN week9departments as b ON a.did = b.did
           GROUP BY depname")
```

```
##            depname COUNT(*)
## 1        Chemistry       11
## 2 Computer Science        8
## 3        Economics        9
## 4      Mathematics        9
## 5       Statistics        5
```

Question 7

- Using LIMIT for table operations helps with faster process of long tables because it can show us the top of the dataset that we want to view without taking time to load everything. It also makes data output neater than showing everything.

Question 8

Part A

```r
# left joining professors and profcourses
table_a <- left_join(x = week9professors, y = week9profcourses, by = "pid")
head(table_a)
```

```
##   pid    fname      email did    cid
## 1  42    Royce    [Royce] MAT MAT131
## 2  42    Royce    [Royce] MAT MAT218
## 3  41 Abhinaba [Abhinaba] ECN ECN111
## 4  41 Abhinaba [Abhinaba] ECN ECN295
## 5  40   Andrea   [Andrea] ECN ECN366
## 6  39   Andrew   [Andrew] CHM CHM221
```

Part B

```r
# left joining table_a and courses
table_a$pid <- as.character(table_a$pid)
table_b <- left_join(x = table_a, y = week9courses, by = "cid")
head(table_b)
```

```
##   pid    fname      email did.x    cid                       cname did.y
## 1  42    Royce    [Royce]   MAT MAT131                   Calculus I   MAT
## 2  42    Royce    [Royce]   MAT MAT218    Elementary Number Theory   MAT
## 3  41 Abhinaba [Abhinaba]   ECN ECN111   Introduction to Economics   ECN
## 4  41 Abhinaba [Abhinaba]   ECN ECN295    ST: Behavioral Economics   ECN
## 5  40   Andrea   [Andrea]   ECN ECN366 Seminar in Health Economics   ECN
## 6  39   Andrew   [Andrew]   CHM CHM221   Organic Chemistry I w/lab   CHM
```

Part C

```r
# left joining table_b and departments using the foreign key from professors
table_c <- left_join(x = table_b, y = week9departments, by = c("did.x" = "did")) %>%
  subset(., select = c("pid", "cname", "depname"))
head(table_c)
```

```
##   pid                       cname     depname
## 1  42                  Calculus I Mathematics
## 2  42    Elementary Number Theory Mathematics
## 3  41   Introduction to Economics   Economics
## 4  41    ST: Behavioral Economics   Economics
## 5  40 Seminar in Health Economics   Economics
## 6  39   Organic Chemistry I w/lab   Chemistry
```

```r
# Sort by depname (ascending)
table_c %>% arrange(depname) %>% head(5)
```

```
##   pid                      cname   depname
## 1  39  Organic Chemistry I w/lab Chemistry
## 2  39   ST: Adv NMR Spectroscopy Chemistry
## 3  33 Physical Chemistry I w/lab Chemistry
## 4  31 Physical Chemistry I w/lab Chemistry
## 5  30    General Chemistry w/lab Chemistry
```

```r
# Sort by pid (descending)
table_c %>% arrange(desc(as.numeric(pid))) %>% head(5)
```

```
##   pid                       cname     depname
## 1  42                  Calculus I Mathematics
## 2  42    Elementary Number Theory Mathematics
## 3  41   Introduction to Economics   Economics
## 4  41    ST: Behavioral Economics   Economics
## 5  40 Seminar in Health Economics   Economics
```

```r
# Sort by cname (ascending)
table_c %>% arrange(cname) %>% head(5)
```

```
##   pid                       cname          depname
## 1  29     Analysis of Algorithms Computer Science
## 2   3     Analysis of Algorithms       Statistics
## 3  15 Analytical Chemistry w/Lab        Chemistry
## 4  13 Analytical Chemistry w/Lab        Chemistry
## 5  34         Applied Statistics       Statistics
```

Part D

```r
# left joining table_b and departments using the foreign key from courses
table_d <- left_join(x = table_b, y = week9departments, by = c("did.y" = "did")) %>%
  subset(., select = c("pid", "cname", "depname"))
head(table_d)
```

```
##   pid                       cname     depname
## 1  42                  Calculus I Mathematics
## 2  42    Elementary Number Theory Mathematics
## 3  41   Introduction to Economics   Economics
## 4  41    ST: Behavioral Economics   Economics
## 5  40 Seminar in Health Economics   Economics
## 6  39   Organic Chemistry I w/lab   Chemistry
```

```r
# Sort by depname (ascending)
table_d %>% arrange(depname) %>% head(5)
```

```
##   pid                       cname   depname
## 1  39  Organic Chemistry I w/lab Chemistry
## 2  39   ST: Adv NMR Spectroscopy Chemistry
## 3  33 Physical Chemistry I w/lab Chemistry
## 4  31 Physical Chemistry I w/lab Chemistry
## 5  30    General Chemistry w/lab Chemistry
```

```r
# Sort by pid (descending)
table_d %>% arrange(desc(as.numeric(pid))) %>% head(5)
```

```
##   pid                       cname     depname
## 1  42                  Calculus I Mathematics
## 2  42    Elementary Number Theory Mathematics
## 3  41   Introduction to Economics   Economics
## 4  41    ST: Behavioral Economics   Economics
## 5  40 Seminar in Health Economics   Economics
```

```r
# Sort by cname (ascending)
table_d %>% arrange(cname) %>% head(5)
```

```
##   pid                      cname          depname
## 1  29     Analysis of Algorithms Computer Science
## 2   3     Analysis of Algorithms Computer Science
## 3  15 Analytical Chemistry w/Lab        Chemistry
## 4  13 Analytical Chemistry w/Lab        Chemistry
## 5  34         Applied Statistics       Statistics
```

Part E

- These tables are not the same. When filtering the data corresponding to Professor Rebelsky(3), Analysis of Algorithms is in the statistics department when using foreign key from "professors", and it is in the CS department when using foreign key from "courses". Both tables are correct, but the one is showing the department name where the professor is in, while the other is showing the department name that the course is in.

```r
filter(table_c, pid == 3)
```

```
##   pid                        cname    depname
## 1   3 Introduction to Data Science Statistics
## 2   3       Analysis of Algorithms Statistics
```

```r
filter(table_d, pid == 3)
```

```
##   pid                        cname          depname
## 1   3 Introduction to Data Science       Statistics
## 2   3       Analysis of Algorithms Computer Science
```

Question 9

- I seem to get the same answer as question 8 on most parts, except for a slight difference in part b. When left joining the table from part A and courses dataset, left_join in r distinguishes between different foreign keys, using did.x for the did in "professors" and did.y for the did in "courses", while SQL LEFT JOIN did not distinguish them and have them both as did. The joined dataset in part b is also ordered differently in r and SQL, but they contain the same information.

Part A

```r
# left joining professors and profcourses
dbGetQuery(mydb, "SELECT a.*, b.cid FROM week9professors as a
           LEFT JOIN week9profcourses as b ON a.pid = b.pid
           LIMIT 5")
```

```
##   pid    fname      email did    cid
## 1  42    Royce    [Royce] MAT MAT131
## 2  42    Royce    [Royce] MAT MAT218
## 3  41 Abhinaba [Abhinaba] ECN ECN111
## 4  41 Abhinaba [Abhinaba] ECN ECN295
## 5  40   Andrea   [Andrea] ECN ECN366
```

Part B

```r
# left joining the table from Part A and courses
dbGetQuery(mydb, "SELECT a.*, b.cid, c.cname, c.did FROM week9professors as a
```

```
           LEFT JOIN week9profcourses as b ON a.pid = b.pid
           LEFT JOIN week9courses as c ON b.cid = c.cid
           LIMIT 5")
```

```
##   pid     fname       email did    cid                      cname did
## 1  30 Elizabeth [Elizabeth] CHM CHM129    General Chemistry w/lab CHM
## 2  17    Maisha    [Maisha] CHM CHM129    General Chemistry w/lab CHM
## 3   8  Rajendra  [Rajendra] CHM CHM129    General Chemistry w/lab CHM
## 4   5   Stephen   [Stephen] CHM CHM129    General Chemistry w/lab CHM
## 5  15      Mark      [Mark] CHM CHM210 Analytical Chemistry w/Lab CHM
```

Part C

- Left joining the table from Part B and departments using the foreign key from professors

```
# Sort by depname (ascending)

dbGetQuery(mydb, "SELECT a.pid, c.cname, d.depname FROM week9professors as a
           LEFT JOIN week9profcourses as b ON a.pid = b.pid
           LEFT JOIN week9courses as c ON b.cid = c.cid
           LEFT JOIN week9departments as d ON a.did = d.did
           ORDER BY depname
           LIMIT 5")
```

```
##   pid                      cname   depname
## 1  13 Analytical Chemistry w/Lab Chemistry
## 2  30    General Chemistry w/lab Chemistry
## 3  33 Physical Chemistry I w/lab Chemistry
## 4  30  Organic Chemistry I w/lab Chemistry
## 5   8    General Chemistry w/lab Chemistry
```

```
# Sort by pid (descending)

dbGetQuery(mydb, "SELECT a.pid, c.cname, d.depname FROM week9professors as a
           LEFT JOIN week9profcourses as b ON a.pid = b.pid
           LEFT JOIN week9courses as c ON b.cid = c.cid
           LEFT JOIN week9departments as d ON a.did = d.did
           ORDER BY pid DESC
           LIMIT 5")
```

```
##   pid                       cname     depname
## 1  42    Elementary Number Theory Mathematics
## 2  42                  Calculus I Mathematics
## 3  41   Introduction to Economics   Economics
## 4  41    ST: Behavioral Economics   Economics
## 5  40 Seminar in Health Economics   Economics
```

```
# Sort by cname (ascending)

dbGetQuery(mydb, "SELECT a.pid, c.cname, d.depname FROM week9professors as a
           LEFT JOIN week9profcourses as b ON a.pid = b.pid
           LEFT JOIN week9courses as c ON b.cid = c.cid
           LEFT JOIN week9departments as d ON a.did = d.did
           ORDER BY cname
           LIMIT 5")
```

```
##   pid                      cname          depname
```

```
## 1   3     Analysis of Algorithms       Statistics
## 2  29     Analysis of Algorithms Computer Science
## 3  13 Analytical Chemistry w/Lab        Chemistry
## 4  15 Analytical Chemistry w/Lab        Chemistry
## 5  12         Applied Statistics       Statistics
```

Part D

- Left joining the table from Part B and departments using the foreign key from courses

```r
# Sort by depname (ascending)

dbGetQuery(mydb, "SELECT a.pid, c.cname, d.depname FROM week9professors as a
           LEFT JOIN week9profcourses as b ON a.pid = b.pid
           LEFT JOIN week9courses as c ON b.cid = c.cid
           LEFT JOIN week9departments as d ON c.did = d.did
           ORDER BY depname
           LIMIT 5")
```

```
##   pid                      cname   depname
## 1  13 Analytical Chemistry w/Lab Chemistry
## 2  30    General Chemistry w/lab Chemistry
## 3  33 Physical Chemistry I w/lab Chemistry
## 4  30  Organic Chemistry I w/lab Chemistry
## 5   8    General Chemistry w/lab Chemistry
```

```r
# Sort by pid (descending)

dbGetQuery(mydb, "SELECT a.pid, c.cname, d.depname FROM week9professors as a
           LEFT JOIN week9profcourses as b ON a.pid = b.pid
           LEFT JOIN week9courses as c ON b.cid = c.cid
           LEFT JOIN week9departments as d ON c.did = d.did
           ORDER BY pid DESC
           LIMIT 5")
```

```
##   pid                       cname     depname
## 1  42                  Calculus I Mathematics
## 2  42    Elementary Number Theory Mathematics
## 3  41   Introduction to Economics   Economics
## 4  41    ST: Behavioral Economics   Economics
## 5  40 Seminar in Health Economics   Economics
```

```r
# Sort by cname (ascending)

dbGetQuery(mydb, "SELECT a.pid, c.cname, d.depname FROM week9professors as a
           LEFT JOIN week9profcourses as b ON a.pid = b.pid
           LEFT JOIN week9courses as c ON b.cid = c.cid
           LEFT JOIN week9departments as d ON c.did = d.did
           ORDER BY cname
           LIMIT 5")
```

```
##   pid                      cname          depname
## 1  29     Analysis of Algorithms Computer Science
## 2   3     Analysis of Algorithms Computer Science
## 3  13 Analytical Chemistry w/Lab        Chemistry
## 4  15 Analytical Chemistry w/Lab        Chemistry
## 5  12         Applied Statistics       Statistics
```

Part E

```r
# foreign key from professors
dbGetQuery(mydb, "SELECT a.pid, c.cname, d.depname FROM week9professors as a
           LEFT JOIN week9profcourses as b ON a.pid = b.pid
           LEFT JOIN week9courses as c ON b.cid = c.cid
           LEFT JOIN week9departments as d ON a.did = d.did
           WHERE a.pid = 3
           LIMIT 5")
```

```
##   pid                        cname    depname
## 1   3       Analysis of Algorithms Statistics
## 2   3 Introduction to Data Science Statistics
```

```r
# foreign key from courses
dbGetQuery(mydb, "SELECT a.pid, c.cname, d.depname FROM week9professors as a
           LEFT JOIN week9profcourses as b ON a.pid = b.pid
           LEFT JOIN week9courses as c ON b.cid = c.cid
           LEFT JOIN week9departments as d ON c.did = d.did
           WHERE a.pid = 3
           LIMIT 5")
```

```
##   pid                        cname          depname
## 1   3       Analysis of Algorithms Computer Science
## 2   3 Introduction to Data Science       Statistics
```

- Same answer as question 8.

```r
dbDisconnect(mydb)
```

```
## [1] TRUE
```